

Business Taxation, Firm Dynamics, and Heterogeneous Policy Impacts*

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August 19, 2026

Abstract

This paper studies how business income taxation affects firm entry, exit, and aggregate outcomes when industries differ in capital intensity. I combine the German IAB Establishment Panel with district-level variation in the local trade tax rate and find that a one-percentage-point increase in the tax rate is associated with a 0.8-percentage-point increase in the annual exit probability among capital-intensive establishments, which is twice as large as that for the full sample. Motivated by empirical findings, I develop a two-sector general equilibrium model with endogenous entry and exit, heterogeneous productivity, and capital accumulation. Incomplete deductibility of capital costs raises the user cost of capital and makes capital-intensive firms more sensitive to business taxation. After calibration to German moments, in the long-run, general equilibrium effects outweigh the direct effect of the tax rate, and a one-percentage-point tax increase reduces business dynamism, with a decline in the capital-intensive sector greater than that in the non-capital-intensive sector. Aggregate employment, consumption, and productivity fall, and economic activity shifts toward non-capital-intensive industries. An illustrative reduction in entry costs increases firm creation and aggregate output. The results demonstrate that capital intensity is important for the incidence of business taxation and its long-run effects on firm dynamics.

JEL classification: D21, E22, E24, H25, L11

Keywords: business taxation, firm dynamics, firm heterogeneity

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1 Introduction

As the economy increasingly relies on industries requiring upfront investment, including manufacturing, semiconductors, clean energy, and AI infrastructure, the macroeconomic consequences of policies affecting capital-intensive firms become increasingly important. Meanwhile, firm dynamics are a key determinant of employment, productivity growth, and the propagation of macroeconomic policies. In particular, firm entry and exit are closely tied to job creation and destruction. Therefore, understanding how policy affects firm entry and exit is central to evaluating the long-run impact on the economy.

One of the important policy tools to stimulate the economy is taxation. Heterogeneous features of firms might induce disproportionate impacts on business dynamism. One important dimension is the level of capital investment they need when they open a business or start production. This paper refers to this feature as the capital intensity of a firm. Specifically, firms in capital-intensive industries require large up-front investments such as in equipment, machinery, and land as well as more capital investment during production. Capital expenditures are typically deducted gradually, so taxable net profits do not include all current period costs. This delayed tax deduction makes the after-tax cost of capital larger considering the time value of money. As a result, taxes can possibly discourage firms with high upfront costs differently, and tax policy may affect who creates jobs and which businesses survive.

This project investigates how corporate taxation affects firm dynamism and the aggregate economy when firms differ in capital intensity. Specifically, how do changes in the cost of capital, induced by local corporate tax rates, disproportionately affect the entry and exit decisions of capital-intensive establishments? How do these differences translate into aggregate impacts on firm distribution, total employment, and aggregate productivity? With the model calibrated to the real-world data, the paper aims to quantify the disproportionate impact arising from differences in capital intensity across firms.

Empirically, the analysis uses German establishment-level survey data. This dataset is used for the following reasons. First, changes in tax rates are rare at the federal level, which makes it hard to investigate their impacts. By contrast, local business tax exhibits greater spatial and temporal variations, creating a natural setting to study how local tax differences shape firms' entry, survival, and hiring decisions. In Germany, for instance, firms face a local trade tax in addition to the federal corporate income tax. The trade tax, levied on business net profits, provides stable local revenue but varies substantially across municipalities—from 7% to over 22%—and fluctuates more over time than the federal rate. Second, aggregate-level datasets have their limitations in capturing important business characteristics. For example, an establishment's legal form is needed to determine which entities are subject to corporate taxation, yet this information is not available in an aggregate dataset. Moreover, establishment-level records allow for more accurate industry classification than aggregated sources. Furthermore, this dataset covers both public and private establishments and is not restricted to large firms. Including private firms is crucial for examining long-run firm dynamics, especially entry and exit.

With the establishment-level data, I estimate a two-way fixed effects model that exploits variation in local trade tax rates across time and districts. Empirical results show that small-sized and capital-intensive establishments experience higher exit rates compared to an average establishment. Specifically, a one-percentage-point increase in the local trade tax rate is associated with a 0.8-percentage-point increase in annual exit probability among capital-intensive establishments, compared with 0.4 percentage points in

the full sample. This 0.8-percentage-point increase is close to a third of the average exit rate and is more than twice the impact on an average establishment.

Guided by the empirical evidence, I develop a general equilibrium model to evaluate the long-run effects of taxation on business dynamics. The model builds upon Hopenhayn (1992), where firms face idiosyncratic productivity shocks, as well as endogenous entry and exit decisions. Following extensions by Clementi and Palazzo (2016) and Sedláček and Sterk (2019), I incorporate investment choices, capital adjustment costs, and tax on business profits. Further, I extend the model for two types of firms that differ in the capital inputs for production (C for capital-intensive firms, and NC for non-capital-intensive firms). This extra feature allows the model to study the differences in capital intensity across firms and how these affect individual firms' decisions as well as the aggregate economy.

Quantitatively, with the model calibrated to match empirical moments, I conduct two counterfactual analyses. First, I examine the impacts of a one-percentage-point increase in the corporate income tax rate on the economy's firm dynamism and aggregate outcomes. An increase in the tax rate reduces firm entries and exits, as well as the total number of firms. It also lowers the average productivity because of less competition. These effects are more pronounced among capital-intensive firms, causing a shift towards non-capital-intensive industries in the economy. The advantage of the model is to investigate the heterogeneous impact across firms with different capital intensities. For instance, there is a substantially larger contraction among C firms, with the firm mass decreasing by 2.3% in the C sector and 0.7% in the NC sector. The masses of entrants and exiters also contract more among C firms. C firms are more exposed to the policy shock because they require larger capital inputs and cannot immediately deduct the full cost of investment. The incomplete deductibility raises their effective user cost of capital. The nonnegative cash-flow constraint for each period of production may further amplify this response. For the aggregate economy, output falls much more in the C sector and employment shifts modestly towards the NC sector. The average firm size rises by roughly 1%, with a larger increase for C firms. As the market becomes more concentrated, average productivity falls.

The decrease in exit rate in the quantitative model and the increase in exit rate in the empirical section unveil the distinctions between direct and general equilibrium effects of a tax rate increase. Holding prices fixed, an increase in the tax rate raises exit rates, particularly among C firms, which is consistent with what is observed in the empirical motivations. In general equilibrium, price adjustment more than offsets the direct effect, causing long-run entry and exit rates to fall. This reconciles the positive short-run empirical exit response with the negative long-run response in the model.

In another counterfactual analysis, motivated by declining artificial intelligence costs, I consider an illustrative 2.3% reduction in entry costs. This increases firm mass, output, and average productivity. Productivity rises because of greater entry and stronger competition in the market.

Literature Review Most of the existing work studies average tax effects or other dimensions of firm heterogeneity. However, little is known about how capital intensity shapes the long-run effects of corporate taxation on firm entry, exit, and aggregate outcomes. The paper contributes to the literature in the following ways.

The paper contributes to the literature on firm dynamics. The workhorse model for this strand of literature was first introduced in Hopenhayn (1992). Following that, Bartelsman et al. (2013) and Clementi and Palazzo (2016) integrate capital investment

and adjustment costs into this framework. Our work is closely related to the work by Sedláček and Sterk (2019), which investigates the impact of tax reform with a corporate income tax cut and a change of expensing regimes on long-term firm dynamics and aggregate economies. The transmission channel includes the direct impact, which is the impact on the incumbent firm when holding the wage constant, and an indirect channel, which evaluates the impact on incumbent firms from changes in equilibrium wages and the total number of firms. Different from their paper, which compares different investment expensing schedules, this paper compares the long-run impact on business dynamics coming from different capital intensities across industries. Different from Sedláček and Sterk (2019), Clementi and Palazzo (2016) has a time-to-build capital and focuses more on the propagation impact of aggregate shocks through firm entry and exit dynamics. My model’s contribution is to add heterogeneity in capital intensities across industries. The heterogeneity matters as large up-front cost differences and expensing schedule differences across labor and capital costs are important in affecting the extensive and intensive margins of firms and aggregate productivity.

The paper is also related to the literature on tax policy and labor demand. Giroud and Rauh (2019) utilize micro-level multi-state firm data to study the impact of state-level corporate tax on employment and a number of establishments. They find that employment and the number of establishments have tax elasticities of -0.4 and -0.5, respectively. The authors also shed light on sector heterogeneity and find that more localized firms or highly labor-intensive industries respond more to taxation changes than the average. Other similar studies include Ljungqvist and Smolyansky (2014). Inspired by the empirical studies, Berger et al. (2022) construct a general equilibrium model of oligopsony to show that large firms will have a lower labor share of income, resulting in a weaker employment reduction with labor demand shocks, such as with state corporate tax changes. The authors also argue that reduced-form estimates in the literature normally underestimate the results.

Apart from the impacts on the labor market, other papers study the impact of tax policy on investment. Crawford and Markarian (2024) use the difference-in-differences approach with public firm data from the U.S. and Canadian firms and show that the Tax Cut and Job Act of 2017 (TCJA) increases firm investment for American firms. In particular, capital-intensive firms and financially constrained firms increase the most which the authors attribute to changes in the expensing schedule provision in TCJA. Other papers mainly focus on accelerated depreciation policies¹. Much of the literature on accelerated depreciation tax policy focuses on the impact on investment growth as in Zwick and Mahon (2017). Recently, more papers have studied the policy’s impact on the labor market. Curtis et al. (2021) shows that accelerated depreciation policy increases employment in the manufacturing sector with both empirical and structural evidence. Garrett et al. (2020) argues that the impact of the bonus depreciation policy differs by asset type, depending on the duration of the capital.

The remainder of the paper is organized as follows. Section 2 presents the empirical motivation of the paper. Section 3 uses a toy model to illustrate the key mechanism proposed in the paper. The full model is introduced in Section 4. Section 5 describes the estimation of the model parameters and presents counterfactual analysis. Section 6

¹The accelerated depreciation policy is a tax stimulus that accelerates the deduction schedule of investment costs from firms’ taxable income. This policy alters the timing of depreciation, providing incentives for firms as it increases the present discounted value of their cumulative deductions, thus lowering the tax.

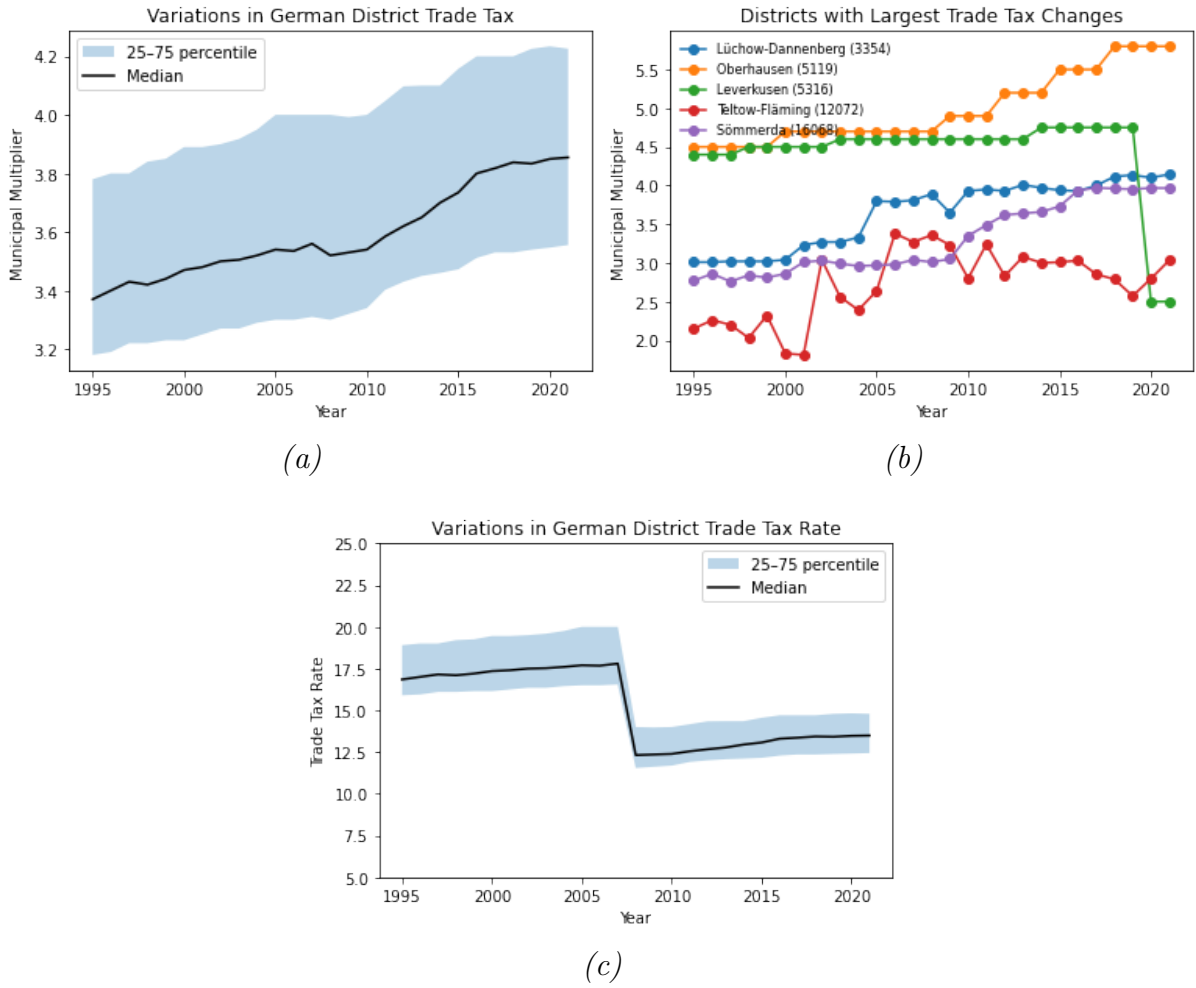
concludes.

2 Empirical Motivations

2.1 Tax system in Germany

The corporate tax rate at the national level is largely stable over time. By contrast, there are more changes in corporate tax rates at the local level. This empirical section utilizes the rich variations in German local corporate tax rates to study their impacts on business dynamics.

Figure 1: German district-level average trade tax rate over time



The business tax system in Germany consists of a federal-level corporate income tax and a municipality-level trade tax. The federal-level corporate income tax includes a uniform tax rate and is subject to a solidarity surcharge tax. Since 2001, there has been only one reform that changed the federal tax rate from 25% to 15% in 2008, and the solidarity surcharge tax rate has remained at 5.5% since 1998. Notice that only corporations are subject to the corporate income tax, and the profits for partnerships and sole proprietorships are passed through and taxed at the individual income tax rate of the owners. On the other hand, trade tax in Germany applies to all commercial

businesses regardless of legal form. The trade tax rate is regulated by a base rate set by the federal government and a scaling factor (municipal multiplier) determined by each municipality, which is the smallest local government unit in Germany.

Figure 1 shows the variations of the average trade tax multipliers at the district level ². Figure 1(a) shows the distribution of the multipliers in the past two decades. Over time, the median of the multiplier increases from about 3.4 to 3.8, and there are large variations across districts. Figure 1(b) shows examples of trade tax multiplier changes for districts with larger changes. In addition, the base rate decreased from 5% to 3.5% in 2008, which translates to a decrease in the median trade tax rate from 16.85% to 13.48% over time (as shown in Figure 1(c)).

The depreciation schedule is also an important factor affecting firm dynamics. In Germany, the depreciation schedule is regulated at the federal level by a standard depreciation table. The depreciation table for general-purpose fixed assets (abbreviated to "AfA Table AV") shows the standard useful life (Section 7, Paragraph 1, Sentence 2 of the Income Tax Act) for fixed assets that are not used for a specific industry. The "AfA Table AV" applies to all fixed assets acquired or manufactured after December 31, 2000.

2.2 Data description

The main dataset used in the paper is the IAB Establishment Panel (IAB-BP³), which is an annual representative employer survey database and serves as the central source for German labor market demand analysis. Since 1996, the data cover both West Germany and East Germany. To better capture the entry and exit information of establishments, I also supplement the IAB-BP data with the entry and exit extension files from the Establishment History Panel (BHP) ⁴, which is a cross-sectional establishment-level dataset created by aggregating the administrative worker-level data.

One of the key measurement challenges is to identify the entry and exit of an establishment. Normally, a similar establishment-level or firm-level dataset only captures the time establishments get surveyed, enter the dataset, or the time a firm goes public, which are not necessarily the time of starting the business. This dataset makes extra efforts to try to classify establishment entries and exits more precisely based on employment and its flows. In the main dataset, an establishment's start-up year is defined as the first year in which it has at least one employee subject to social security contributions on

²The average trade tax rate was sourced from the Regional Database Germany directly via <https://www.regionalstatistik.de/genesis/online>. The paper is looking at the district average trade tax rate changes because the location of establishment is only available at district level for confidentiality reasons.

³Bellmann, Lisa; Gensicke, Miriam; Grau Katrin; Kohaut, Susanne; Möller, Iris; Oberfichtner, Michael; Schwengler, Barbara; Tschersich, Nikolai; Umkehrer, Matthias (2026): "IAB-Betriebspanel (IAB-BP) – Version 9324 v1. Forschungsdatenzentrum (FDZ) der Bundesagentur für Arbeit (BA) im Institut für Arbeitsmarkt- und Berufsforschung (IAB). DOI: 10.5164/IAB.IABBP9324.de.en.v1. Data access was provided via on-site use at the Research Data Centre (FDZ) of the German Federal Employment Agency (BA) at the Institute for Employment Research (IAB) and subsequently via remote data access. Bellmann et al. (2026)

⁴Ganzer, Andreas; Schmucker, Alexandra; Wolter, Stefanie (2024): "Establishment History Panel 1975–2023 (BHP 7523 v1)". Research Data Centre of the Federal Employment Agency (BA) at the Institute for Employment Research (IAB). DOI: 10.5164/IAB.BHP7523.de.en.v1 Data access was provided via remote desktop/on-site use at the Research Data Centre (FDZ) of the German Federal Employment Agency (BA) at the Institute for Employment Research (IAB) and remote execution. Ganzer et al. (2024)

June 30. Then, using the entry and exit extension files, I exclude establishments whose recorded entry type reflects a spin-off or an identification-number change. The decisive factor to determine the entry type is the proportion of workers who were employed in the predecessor and the new establishments. The clean-up of the closure of an establishment is done along the same line. Details on entry/exit type classifications can be found in Hethey-Maier and Schmieder (2013).⁵

I proxy for an industry’s capital intensity using gross fixed capital formation per worker, measured by the gross fixed capital formation over labor input from Eurostat over 2000 to 2020. I classify the NACE Rev. 2 industries as capital intensive (C) if the measure is above the median and those as non-capital intensive (NC) if below the cutoff. Detailed industry classification can be found in the Appendix Table A.1. Local tax rate data and other regional aggregates, such as population, are taken from German regional datasets.

2.3 Empirical methodology

The empirical section tests how establishment-level entry and exit decisions respond to local business tax changes. Do the changes differ across establishment ages and sizes? And are the capital-intensive industries also more policy-sensitive?

The empirical specification for the exit decision is

$$Y_{i,d,s,t} = \beta_0 + \beta_1 \Delta\tau_{d,t} + \beta_2 \Delta\tau_{d,t-1} + \beta_3 \Delta\tau_{d,t-2} + X'_{i,t} \gamma + Z'_{d,t} \rho + \delta_d + \alpha_{s,t} + \varepsilon_{i,d,s,t} \quad (1)$$

, where i indexes establishments, d indexes districts, s indexes sectors, and t indexes years. The dependent variable $Y_{i,d,s,t}$ equals one if establishment i is recorded as exiting in year t , and zero otherwise. The variable $\Delta\tau_{d,t}$ is the district-level trade tax rate percentage point changes, and

$$\Delta\tau_{d,t} = \tau_{d,t} - \tau_{d,t-1}.$$

Equation 1 includes the contemporaneous tax changes and two lags. The vector X_{it} contains establishment-level controls, including an establishment’s legal form, size, and age. Legal forms of establishments include Individual company (sole proprietor), Partnerships, Corporation (GmbH, AG, cooperative), and Non-commercial/public-law/other. Establishments are divided into three size categories based on their number of employees: small establishments with 1 to 19 employees, medium establishments with 20 to 199 employees, and large establishments with at least 200 employees. The vector $Z_{d,t}$ contains time-varying district-level characteristics, such as population, other taxation, such as property taxes, as well as GDP growth.

All specifications include district fixed effects, δ_d , which absorb differences across districts, such as local economic conditions. In specifications pooling all samples, I include sector-by-year fixed effects, $\alpha_{s,t}$, which control for common shocks varying across each sector group over years. In specifications where the samples are restricted to capital-intensive establishments, year fixed effects are controlled.

With respect to the entry decision, empirical data can only observe those who chose to enter the market but not those who intended to but did not due to various reasons.

⁵Entry and exit years occasionally differ between the IAB-BP and BHP files, primarily because the IAB-BP excludes establishments that employ only marginal part-time workers, whereas the BHP includes them. I use the entry and exit years reported in the IAB-BP as the primary measures. I supplement these measures with information from the BHP to identify the type of entry or exit and to fill in values missing from the IAB-BP.

Given this, instead of using the entry dummy as the response variable, I aggregate the establishment-level observations to the district-sector-year level and run a similar specification in Equation (2).

$$Y_{d,s,t} = \alpha + \sum_{\ell=0}^3 \beta_{\ell} \Delta \tau_{d,t-\ell} + Z'_{d,t} \rho + \delta_d + \alpha_{s,t} + \varepsilon_{d,s,t} \quad (2)$$

2.4 Trade tax and establishment dynamics

Table 1 estimates the relationship between changes in local trade tax rates and establishment exit decisions. In Table 1, columns 1-4 present the results for all establishments, establishments in capital-intensive industries, small-sized establishments, as well as small-sized and capital-intensive establishments, respectively. Columns 5-8 report regressions for the same set of samples but include additional control variables, such as local GDP growth and property tax rate changes.

The contemporaneous coefficient of tax rate changes is positive and statistically significant at the 5% level across all specifications. In the full sample, column (1) indicates that a one-percentage-point increase in trade tax rate is associated with a 0.4-percentage-point increase in the annual probability of establishment exit. This correlation doubles for capital-intensive establishments, ranging from 0.8 to 0.9 percentage points, as shown in columns (2) and (4). Notice that the average exit rate in the full sample is about 2.6% in this panel sample. So the observed changes in exit rate associated with tax rate changes are about 15-30% of the average exit rate. Both columns (1) and (2) also indicate that compared to establishments with more than 200 employees, those with less than 20 employees have a statistically significantly higher exit probability. The negative coefficient of age shows that exit probability is declining as establishments mature. Establishments' legal forms are also significantly correlated with exit probabilities. Compared to the non-commercial establishments, the individual company (sole proprietor) is more likely to have a higher exit rate. With more controls in columns (5) to (8), the results still hold. Notice that the impact of Germany's Business Tax Reform Act of 2008 is controlled with the year fixed effect. Appendix B provides a robustness check by excluding the sample in 2008.

Table 2 presents the estimated relationship between changes in the local trade tax rate and establishment entry. The dependent variable in columns (1) and (2) is the establishment entry rate, while that in columns (3) and (4) is the percentage-point change in entry rate. Table 2 shows that an increase in the trade tax rate is associated with a decrease in establishment entry rate or slower growth in entry rate across the specifications. In addition, the point estimates are also larger in absolute value in the capital-intensive subsample than in the full samples. Although these estimates do not reach conventional levels of statistical significance, they can offer suggestive evidence that capital-intensive establishments behave differently than an average establishment with the same level of changes in local business tax. The insignificance of the coefficients might partly reflect the smaller number of observations after aggregating the data from an establishment-level to district-level data sample, and the low frequency of entry events.

Table 1: Impact of trade tax change on exits

	(1) All	(2) Capital intensive	(3) Small	(4) C and Small	(5) All	(6) Capital intensive	(7) Small	(8) C and Small
Tax changes	0.004** (0.002)	0.008** (0.004)	0.004** (0.002)	0.009** (0.004)	0.004** (0.002)	0.008** (0.004)	0.004** (0.002)	0.009** (0.004)
Tax changes, lag 1	0.001 (0.001)	0.002 (0.002)	0.001 (0.002)	0.002 (0.002)	0.001 (0.001)	0.002 (0.002)	0.001 (0.002)	0.002 (0.002)
Tax changes, lag 2	0.004** (0.002)	0.005* (0.002)	0.004** (0.002)	0.005* (0.003)	0.004** (0.002)	0.005* (0.002)	0.004** (0.002)	0.005** (0.003)
1 to 19	0.017*** (0.002)	0.025*** (0.006)			0.017*** (0.002)	0.026*** (0.006)		
20 to 199	0.003 (0.002)	0.007 (0.005)			0.003 (0.002)	0.007 (0.005)		
Age	-0.003*** (0.000)	-0.004*** (0.001)	-0.003*** (0.000)	-0.004*** (0.001)	-0.003*** (0.000)	-0.003*** (0.001)	-0.003*** (0.000)	-0.004*** (0.001)
Age ²	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
Log of Population	-0.039 (0.063)	-0.024 (0.108)	-0.048 (0.069)	-0.019 (0.125)	-0.042 (0.063)	-0.030 (0.108)	-0.051 (0.069)	-0.025 (0.125)
Individual company	0.014*** (0.003)	0.022** (0.008)	0.014*** (0.004)	0.020* (0.011)	0.014*** (0.003)	0.021** (0.009)	0.014*** (0.004)	0.019* (0.011)
Partnership	0.005 (0.005)	0.010 (0.010)	0.0 05 (0.006)	0.008 (0.013)	0.005 (0.005)	0.009 (0.011)	0.005 (0.006)	0.008 (0.013)
Corporation	-0.002 (0.003)	0.007 (0.008)	-0.003 (0.004)	0.003 (0.011)	-0.002 (0.003)	0.007 (0.008)	-0 .003 (0.004)	0.003 (0.011)
Short controls	Yes	Yes	Yes	Yes	No	No	No	No
Long controls	No	No	No	No	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	No	Yes	No	Yes	No	Yes	No
Year x Sector FE	Yes	No	Yes	No	Yes	No	Yes	No
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of Obs.	79,784	32,905	52,935	19,762	77,664	32,029	51,648	19,269
R-Squared	0.023	0.047	0.023	0.051	0.023	0.047	0.023	0.051

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors are clustered at the district level. The sample period from 2008 to 2017. Regressions are adjusted using panel weights.

3 Key mechanism in a simple model

The important assumption that allows the tax rate to influence capital-intensive firms disproportionately is that the capital cost is not expensed immediately. Corporate income tax affects firms' marginal cost of capital but not the unit cost of labor. That is because capital expenditures will be depreciated in a delayed manner, while the labor cost will be deducted immediately. When the corporate income tax rate increases, the delayed expensing of capital costs and the time value of money induce a disproportionate impact on capital-intensive firms. This mechanism is amplified or more influential at the entry stage of firms since they are more financially constrained. This section uses a simple static two-sector firm model to illustrate this. For illustrative purposes, imagine an extreme case where a capital-intensive firm only requires capital and a non-capital-intensive firm only requires labor for production.

3.1 Static firm problem

In a simplified static model, there are two types of firms, where C firms represent capital-intensive firms and NC firms represent non-capital-intensive firms. The value function of C firms can be expressed as follows:

$$V^C(k_{-1}) = \max_k (1 - \tau)[f^C(k) - c_f] - (1 - \lambda\tau)[k - (1 - \delta)k_{-1}] + \beta V^C(k) \quad (3)$$

Table 2: Impact of trade tax change on entries

	Entry rate		Growth in entry rate	
	(1)	(2)	(3)	(4)
	All	Capital intensive	All	Capital intensive
Tax changes	-0.004 (0.004)	-0.006 (0.008)	-0.008 (0.007)	-0.020 (0.015)
Tax changes, lag 1	-0.004 (0.003)	-0.008 (0.007)	0.002 (0.008)	-0.005 (0.015)
Tax changes, lag 2	-0.010 (0.007)	-0.015 (0.011)	-0.005 (0.007)	-0.004 (0.007)
Tax changes, lag 3	-0.009 (0.007)	-0.019 (0.016)	-0.004 (0.005)	-0.012 (0.014)
Year FE	Yes	Yes	Yes	Yes
Sector FE	Yes	No	Yes	No
Year x Sector FE	Yes	No	Yes	No
District FE	Yes	Yes	Yes	Yes
No. of Obs.	7,963	3,943	6,446	3,188
R-Squared	0.093	0.140	0.035	0.049

Notes: Standard errors are clustered at the district level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sample period from 2008 to 2020. Regressions are adjusted using average firm size.

, where $i = k - (1 - \delta)k_{-1}$. Here, λ , the effective expense rate, is a reduced-form way to show the present value of future deductions⁶, such as literal expensing, depreciation allowances, and investment tax credits. Since capital cost is not expensed immediately, λ is strictly smaller than 1.

First-order condition of equation (3) w.r.t. k becomes:

$$(1 - \tau) \frac{\partial f}{\partial k} - (1 - \lambda\tau) + \beta \frac{\partial V^C}{\partial k} = 0$$

$$\frac{\partial f}{\partial k} = \Omega = \left(\frac{1 - \lambda\tau}{1 - \tau} \right) \left(\frac{r + \delta}{1 + r} \right) \quad (4)$$

, where Ω is the user cost of capital, $\beta = \frac{1}{1+r}$.

$$\frac{\partial \Omega}{\partial \tau} = \frac{r + \delta}{1 + r} \cdot \frac{1 - \lambda}{(1 - \tau)^2} > 0, \text{ with } \lambda < 1 \quad (5)$$

The partial derivatives of equation (4) with respect to the tax rate τ yield equation (5), which shows that an increase in the tax rate will increase the user cost of capital when λ is smaller than 1. So, the tax rate will distort C firms' capital choices by increasing the user cost of capital. In addition, as the tax rate increases, the distortion increases.

By contrast, the value function for an NC firm is as follows:

$$V^{NC}(w) = \max_l (1 - \tau)[f^{NC}(l) - c_f] - (1 - \tau)wl$$

And the first-order condition w.r.t. l becomes:

$$(1 - \tau) \frac{\partial f}{\partial l} - (1 - \tau)w = 0$$

$$\frac{\partial f}{\partial l} = w$$

The unit cost of labor does not depend on the tax rate.

⁶As in Barro and Furman (2018) and Sedláček and Sterk (2019)

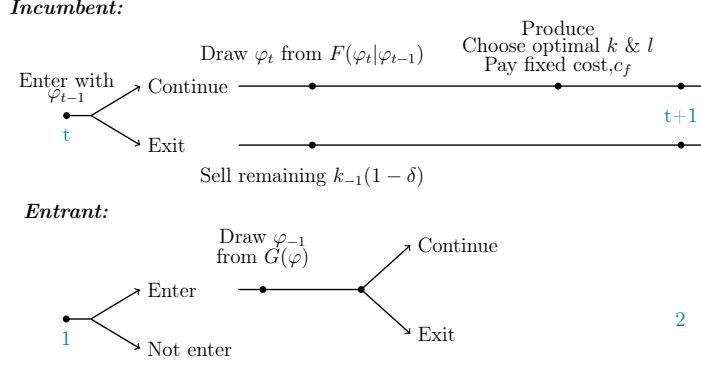


Figure 2: Timeline

4 Two-sector dynamic model

4.1 Model Environment

This paper starts with the classic firm entry-exit model by Hopenhayn (1992) with endogenous entry and exit decisions by firms. Similar to Clementi and Palazzo (2016) and Sedláček and Sterk (2019), this model also incorporates capital investment and corporate income tax. This model's contribution is to add heterogeneity in capital intensities across industries. There are two types of competitive firms: capital-intensive (C) firms and non-capital-intensive (NC) ones. They differ in the amount of capital they need for a unit of production, and the produced goods are heterogeneous as well. The representative household provides labor, gains firms' profits, and consumes the produced goods. Prices of goods, wages, and mass measures are determined endogenously with free-entry conditions of firms and clearing of aggregate demand and supply of goods.

4.2 Timing of firms

Time is discrete and continues forever. The timeline of the model is illustrated in Figure 2. For incumbent firms, when making continue or exit decisions, they do not know the realized productivity level at the current period. If a firm chooses to continue to produce, the productivity level is realized and they start to produce by choosing optimal levels of capital and labor. They also need to pay a fixed operation cost in each period of production. If the firm chooses to exit, it needs to sell the remaining underappreciated part of capital from the previous period. For entrants, they are free to enter the market after paying an entry cost. After entering, an entrant draws its initial lagged value of productivity from a distribution and starts to behave as an incumbent.

4.3 Firm's problem

4.3.1 Production

There are two types of firms, $i = \{C, NC\}$, which have different capital share parameters, α^C and α^{NC} . Firms use both capital and labor as production inputs, and C firms use capital more intensively than NC firms. Productivity shock $\varphi \in S \equiv [0, 1]$ follows a Markov process independent across firms with conditional distribution $F(\varphi|\varphi_{-1})$. The distribution is strictly increasing in φ , meaning a firm with high productivity today is more likely to have high productivity next period. Firms behave competitively, taking prices in the output and input markets as given. Entry costs, operating costs, and investment are denominated in units of the firm's own sector output.

Firm's production function is

$$f^i(\varphi, k, n) = \varphi(k^\alpha l^{1-\alpha})^\theta$$

4.3.2 Incumbent's problem

Given a vector for prices $z^i = \{p^i, w^i\}$, incumbent firms need to make exit decisions:

$$V^i(\varphi_{-1}, z^i, k_{-1}) = \max \left\{ V_c^i(\varphi_{-1}, z^i, k_{-1}), V_x^i(k_{-1}) \right\} \quad (6)$$

If they choose to continue, firms need to maximize the present value of after-tax profits. In addition, firms are subject to a nonnegative cash-flow constraint ($\pi^i \geq 0$), which requires current revenues to cover production costs, such as operating and investment costs. This restriction rules out external borrowing. In addition, the existence of fixed operation costs provides a mechanism for firm exit. If there were no fixed costs, then the firm would not exit and could just wait for a possible higher productivity realization.

If an incumbent firm chooses to continue, the value function is:

$$V_c^i(\varphi_{-1}, z^i, k_{-1}) = \max_{l, k: \pi^i(\varphi, k, l; k_{-1}) \geq 0} \pi^i(\varphi, k, l; k_{-1}) + \beta \int V(\varphi, z^i, k) dF(\varphi | \varphi_{-1}) \quad (7)$$

,where

$$\pi^i = (1 - \tau)[p^i y^i(\varphi, k, l) - wl - c_f - g(k_{-1}, k)] - (1 - \lambda\tau)x$$

and investment is $x = k - (1 - \delta)k_{-1}$. Capital is subject to adjustment cost $g(k_{-1}, k)$, which follows $g(k_{-1}, k) = \zeta_0(x)k_{-1} + \frac{\zeta_1}{2}(\frac{x}{k_{-1}})^2 k_{-1}$ as in Clementi and Palazzo (2016). If there is new investment, x , the adjustment cost contains a non-convex part and a convex part. The fixed cost parameter $\zeta_0(x) = 0$ if $x = 0$ and $\zeta_0(x) > 0$ if $x \neq 0$. The gradual expensing of capital cost is captured with an effective expensing rate, λ , which is a reduced-form way to show the present value of future deductions (also in Barro and Furman (2018) and Sedláček and Sterk (2019)).

If they choose to exit, they will sell the undepreciated part of capital from last period and pay for the adjustment cost of selling the assets. The value function for exit becomes:

$$V_x^i = k_{-1}(1 - \delta) - g(k_{-1}, k)(1 - \tau) \quad (8)$$

4.3.3 Entrant's problem

The potential entrants are exogenously assigned to work in the C or NC industry. Entrants need to pay the entry cost first, and then the productivity is revealed. The model assumes that the entry cost is not tax-deductible, as it is a cost associated with red tape and the business plan development, following the same assumption as in Sedláček and Sterk (2019). The initial lagged productivity is drawn from a lognormal distribution, $G(\varphi)$. The value function for the entrants is:

$$V_e^i(\varphi, z^i, 0) = \max \left\{ 0, -c_e^i + \int V^i(\varphi_{-1}, z^i, 0) dG(\varphi_{-1}) \right\} \quad (9)$$

So the free-entry condition requires that

$$c_e^i = \int V^i(\varphi_{-1}, z^i, 0) dG(\varphi_{-1})$$

After entry, in the same period, entrants behave the same as an incumbent. So, in this environment, there are entrants that enter and exit in the same year when they can not sustain a non-negative after-tax profit in the first period of production.

4.4 Other agents

Representative households consume according to the standard CES preferences over two types of goods C^C and C^{NC} . Labor is elastic and perfectly mobile across sectors. $\sigma = 1/(1 - \rho)$ is the elasticity of substitution. The aggregate consumption good is

$$C \equiv \left[\omega (C^C)^{\frac{\sigma-1}{\sigma}} + (1 - \omega) (C^{NC})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

and the CES price index P is $P = [\omega (p^C)^{1-\sigma} + (1 - \omega) (p^{NC})^{1-\sigma}]^{\frac{1}{1-\sigma}}$. ω is the weight on the capital-intensive good. σ is the elasticity of substitution between the two sectoral goods. The household's utility function is $U = u(C) - \nu L$.

The government is assumed to run a balanced budget by collecting taxes from firms and rebating them to households in a lump-sum way.

4.5 Equilibrium

The competitive equilibrium consists of prices $\{z^{i,*}\}$, mass of entrants $\{m^{i,*}\}$, exit threshold $\{\varphi^{i,*}\}$, distribution of firms $\{\mu(\varphi)^{i,*}\}$ such that

- With given prices, $\{z^{i,*}\}$, individual firms decide on entry or exit from the market, as well as the required optimal labor, $l^{i,*}$, and capital, $k^{i,*}$, inputs for production.
- Households' problem determines aggregate demand for consumption goods (C^i) and labor supply (L).
- Firms' distribution is stationary: $\mu_t^i(k^*, \varphi_t) = \int_{\varphi \geq \varphi^{i,*}} F(\varphi_t | \varphi_{t-1}) \mu_{t-1}^i(d\varphi) + m_t^i G(\varphi)$
- Goods markets for the two types of goods clear with $Y^i = C^i + I^i + G^i + \pi^i$, where I includes aggregate investments, adjustment cost, entry cost, and fixed cost of operation. $Y^{i,s} = \mu^i * f^i$. G^i is the public goods, and π^i is the after-tax firm profits.
- Labor market clears: $L = \mu^C * l^C + \mu^{NC} * l^{NC}$

4.6 Computation steps

1. Solve the individual firm's problem. Given prices, the firm decides on capital and labor inputs, as well as exit decisions. Then, solve for the endogenous prices with the free-entry condition.
2. Guess entrants' mass, and iterate to get the stationary firm distribution with entry and exit decisions.
3. With stationary firm mass, calculate household's demand. Specifically, demand functions for two goods depend on household income, prices of goods, as well as the CES utility parameters.
4. Check for goods market clearing conditions.
5. Update entrant mass and return to step 2.

5 Quantitative Results

5.1 Calibration

Parameters listed in Table 3 are taken directly from the data or from the literature. The annual discount factor, β , is set so that the annual risk-free return is 2%, reflecting the average policy rate in Germany since the 2000s. θ , the span of control, governs the returns to scale of the industries. In the literature, it is widely agreed that micro-level returns to scale (RTS) is more likely to be decreasing. Many estimations of RTS are based on the U.S. setting (Basu and Fernald (1997)). In McAdam et al. (2024), the authors estimated the RTS for five countries in the Euro area⁷, and the RTS estimates range from 0.74 to 1.22. Sedláček and Sterk (2019) and Clementi and Palazzo (2016) study firm dynamics follow similar studies as this paper does but calibrated in the setting for the U.S., the span of control parameters in their papers are 0.9 and 0.8, respectively. Setting the span of control to be 0.8 lies in the reasonable range according to the literature. The depreciation rate, δ , and constant elasticity of substitution, σ are taken from the literature. λ is estimated using broad asset categories following the standard German depreciation table weighted by the asset shares in 2010. Notably, economic depreciation is not incorporated, so it might be even lower once considered. Prices of goods produced by non-capital-intensive firms are normalized to 1. The tax rate is calculated from German administrative data and includes federal-level corporate income tax, solidarity surtax, and municipal trade tax from 2008 to 2018.

Table 3: Assigned parameters

Parameter	Description	Value/Source
β	Annual discount factor	0.9804
θ	Span of control	0.8
δ	Depreciation rate	0.1, Clementi and Palazzo (2016)
λ	Imperfect deductibility of capital costs	0.87
p^{NC}	Price of NC goods	1
σ	Constant elasticity of substitution	2.9, Broda and Weinstein (2006)
τ	Corporate tax rate	0.2955
ν	Disutility of labor	0.01, Sedláček and Sterk (2019)

Parameters listed in Table 4 are jointly calibrated to align the model with key features of the data. The main datasets used are Eurostat data from 2000 to 2020, and IAB data for the period 2000-2020. For moments calculated from survey data, sampling weights are properly applied. Operation cost is calibrated by targeting the average exit rate of incumbent firms in the data. Entry costs for the two types of firms are targeted to match the average entry rate across all the establishments and the exit rate of new entrants for C firms. The exit rate of new entrants is defined as the share of entrants that enter and exit within the same year. Production parameters, α_C and α_{NC} , are calibrated by targeting the capital share of C and NC industries using Eurostat data. Adjustment cost parameters, ζ_0 and ζ_1 , are calibrated by targeting moments related to investment behaviors estimated in the IAB data. Investment rate is defined as the ratio of total investment in the previous year to business volume in that year. The inaction rate is defined as the fraction of periods in which the investment rate is less than 1%, following definitions commonly used in the literature, such as Clementi and Palazzo (2016). The persistence of productivity shock for incumbents is calibrated by targeting the ratio of employment of large firms (those at the 90th percentile) to that of the median firm. Finally, households' CES preference weight is calibrated to match the relative output sizes of the C and NC sectors, using Eurostat data.

⁷Countries include Belgium, France, Italy, Portugal, and Spain.

Table 4: Jointly calibrated parameter values

Parameter	Description	Estimate	Targeted Moment	Data/Source	Model
c_f	Operation cost	0.3452	Average exit rate for incumbents	0.0191, IAB	0.0206
c_e^C	Entry cost, C firms	0.4433	Exit rate of entrants, C	0.0895, IAB	0.0913
c_e^{NC}	Entry cost, NC firms	1.6691	Entry rate	0.0232, IAB	0.0206
α_C	Production parameter, C	0.3786	Capital share, C	0.5029, Eurostat	0.5029
α_{NC}	Production parameter, NC	0.1569	Capital share, NC	0.3255, Eurostat	0.3255
ζ_0	Adjustment cost, non-convex part	0.0016	Inaction rate	0.1940, IAB	0.1685
ζ_1	Adjustment cost, convex part	0.9465	Average investment rate	0.1113, IAB	0.1392
ρ_s	Persistence of incumbents' productivity shocks	0.7159	Relative size of 90th/50th firm	4.4, IAB	4.9695
σ_s	S.D. of incumbents' productivity shocks	0.2480	S.D. of investment rate	0.4643, IAB	0.4525
ω_C	CES preference weight	0.5701	Relative size of C/NC sector	1.0737, Eurostat	1.0738

5.2 Model Dynamics

Figure 3 shows firms' capital decisions. As expected, as the productivity level goes up, C firms will increase their asset levels, as shown in Figure 3 (a). The lumpy investment behavior is consistent with non-convex adjustment cost assumption in the model. Figure 3 (b) presents asset levels of continuing firms in both types. The dashed lines indicate the decisions of NC firms and the solid lines are those for the C firms. As expected, C firms invest more than NC firms given the same levels of current-period assets and productivity. In addition, for both firm types, investment increases as productivity goes up.

Figure 3: Choices of capital by firm types

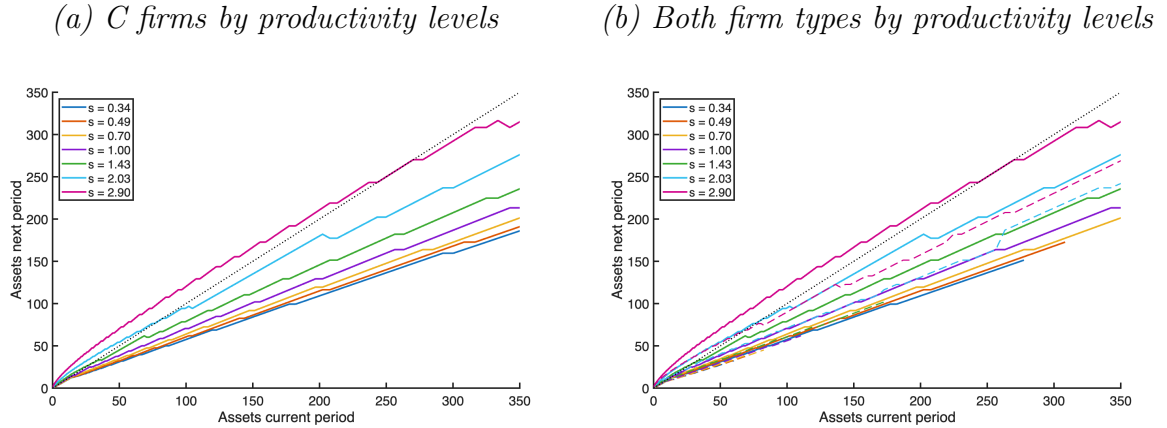


Figure 4 and Figure 5 present firm distributions across capital and employment levels. Figure 4 shows that C firms optimally choose higher capital levels than NC firms. In addition, C firms are more spread out across different capital levels than the NC firms, and are right-skewed, with some high-capital firms. Figure 5 shows that both types of firms are strongly right-skewed, with a large number of firms operating at relatively low employment levels. In addition, the peak of the NC firms' mass is higher than that of C firms, and the corresponding employment level is lower. These observations indicate that NC firms are predominantly smaller in scale and more concentrated in distribution in terms of employment levels. By contrast, C firms' distributions are more spread out.

Table 5 displays firm compositions in the baseline economy. For the economy as a whole, 88% of all the firms continue operating. Out of all the exits, the majority (close to 85%) of them come from entrants. This indicates that it is hard for a new entrant to survive in this economy with the nonnegative profit assumption in the model. But once they survive and start producing, their exit probability decreases substantially. This pattern is more obvious for C firms when comparing results in columns (1) and (2). More than 90% of the capital-intensive firms continue to produce and only 0.1% of the exits come from incumbent C firms. That is

Figure 4: Firm distribution by capital levels

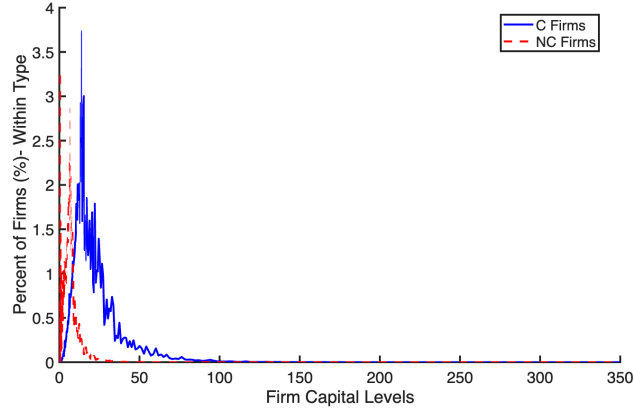
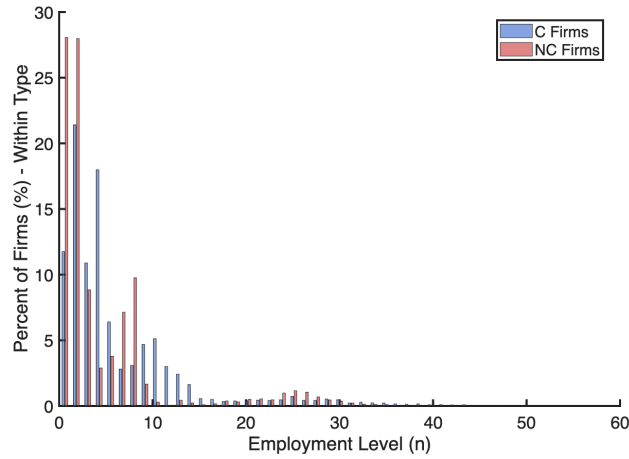


Figure 5: Firm distribution by employment levels



because the current period nonnegative profit assumption for producing is more restrictive for C firms, as they require a larger up-front capital cost. This raises the productivity threshold for entry among C firms. Because entry into the C sector is more selective, firms that enter and produce are, on average, more productive and therefore less likely to exit than firms in the NC sector.

Table 5: Firm Composition: Baseline (% of total firms)

	C Sector	NC Sector	Total
Continuing firms (incl. surviving entrants)	91.56	86.48	88.42
Exiters (all)	8.44	13.52	11.58
Exiting entrants	8.36	10.62	9.76
Exiting incumbents	0.08	2.90	1.82
Entrants	8.44	13.52	11.58

5.3 Counterfactual: increase of tax rate by 1 pp

This section investigates the impact of a one-percentage-point increase in the tax rate, and its impacts on firm dynamism and the general equilibrium of the calibrated economy.

The changes in firm dynamics are summarized in Table 6 and show a decrease in business dynamism with the increase in the tax rate. From the columns for the C sector and the NC sector, it illustrates that although C firms and NC firms experience the same direction of changes in all the measures, the magnitude of the changes varies across sectors. First of all, the drop in the C sector's total firm mass is approximately three times as large as that in the NC sector. This is consistent with what is shown in the illustrative model, which shows that the user cost of capital increases with an increase in the tax rate. That is why the C firms experience a large decrease in the total mass of firms. Moreover, the decrease in exit rate is larger for C firms, and the decrease in entry rate is larger for NC firms. That is because the productivity threshold for entry was already relatively high for C firms, as they need more up-front investment to begin producing. As a result, fewer firms are residing near the cutoff threshold. By contrast, there are more NC firms near the entry margin, and the increase in tax rate can have a larger impact on the entry rate.

The total number of firms also decreases, which is a result of decreases in both entrants and exiters. Panel B in Table 6 shows the long-run steady-state impact of the tax rate increase on the rate-related metrics. There are decreases in entry and exit rates for total firms, as well as for firms in each sector. This might seem counterintuitive and is inconsistent with the empirical motivations, where exit rates increase and entry rates decrease after a tax rate increase. The differences in the entry and exit rate change directions between the quantitative model results and the empirical motivation are due to the differences in the horizon we are looking at. In the empirical section, the positive correlation between tax rate and exit is a contemporaneous direct effect. However, in the quantitative model, the changes shown in Table 6 are between two steady states, which include general equilibrium price adjustments with the policy changes over a long-term horizon.

For the direct tax impact, holding prices fixed, the unit cost of capital will increase as the tax rate increases, which encourages more firms to exit and discourages entries. However, in the new steady states, the general equilibrium effect will raise goods prices, resulting in more entries with increasing profits. This general equilibrium channel will decrease exits and increase entries. So, the dynamics shown in Table 6 are combinations of the two effects. With the price channel dominating total impacts for exit decisions, the total long-run dynamics show a decrease in both rates. The proposed channels can be further indicated by the price changes in Table 7.

Table 6: Entry and Exit (Policy 1 vs. Baseline)

	C Sector	NC Sector	Total
<i>Panel A: Masses (% change)</i>			
Total firms	-2.287	-0.741	-1.331
Continuing firms	-2.040	-0.554	-1.141
Entrants	-4.967	-1.936	-2.779
Exiters	-4.967	-1.936	-2.779
<i>Panel B: Rates (pp change)</i>			
Entry rate	-0.002	-0.035	-0.012
Exit rate	-0.231	-0.163	-0.170
Entrant exit rate	-0.229	-0.128	-0.158

Table 7 shows the changes in aggregate variables and prices compared to the baseline case. With a higher tax rate, firms become less profitable, and C firms suffer more than NC firms. With the equilibrium effect, the relative price of C goods to NC goods will go up, and wages go down to encourage more firms to enter the market. In addition, with the tax increase, real output for both firm types goes down, and the aggregate CES consumption goes down as well. There is a decrease in employment in C sectors and a slight increase in employment for the NC sector. This indicates that there is a slight shift from C sectors to NC sectors in the economy

with a higher tax rate. This also shows that C firms suffer more from the same tax rate change in both sectors.

Table 7: *Equilibrium Outcomes (Policy 1 vs. Baseline, % change)*

	C Sector	NC Sector	Total
Relative price (p_C/p_{NC})	+0.334		
Entrant mass	-4.967	-1.936	-2.779
Output	-1.003	-0.062	
Nominal GDP	-0.672	-0.062	-0.378
Average firm size	+1.593	+0.690	+1.015
Avg productivity, all operating firms	-0.011	-0.080	-0.030
Avg productivity, incumbents (operating)	-0.009	-0.070	-0.025
Employment	-0.479	+0.133	-0.138
Consumption	-0.845	+0.118	
Aggregate consumption (CES)	-0.358		
Wage	-0.194		
Real wage	-0.358		

This exercise also sheds light on implications for aggregate economic productivity and firm distributions. Average firm size increases in both sectors, with the increase in the C sector more than twice that in the NC sector. Together with the decline in firm mass, this pattern indicates that each sector consists of fewer but larger firms following the tax increase and is consistent with greater market concentration. However, this concentrated firm distribution is accompanied by a less productive economy. Average productivity declines among all operating firms. The decline reflects competition in the market. Lower entry reduces competition pressure and allows relatively unproductive incumbents to remain active. As a result, the composition of operating firms shifts towards lower-productivity firms. Similar results have also been discussed in an empirical work by Aw et al. (2003).

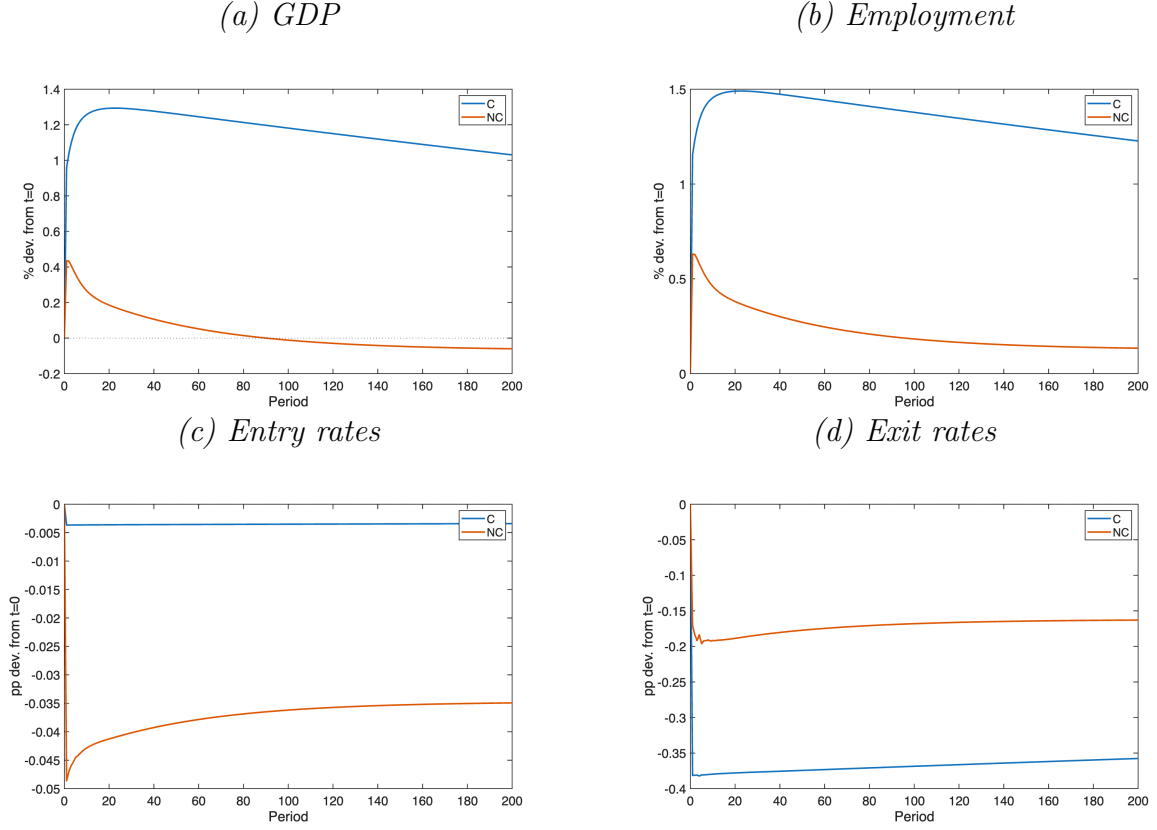
Figure 6 illustrates the firm distributional transition path from the baseline economy to the new steady states, assuming firms quickly adjust to the new steady states' prices and policy functions. For both types of firms, GDP and employment increase first and gradually decrease as firm distribution becomes stationary. Since the relative price of C goods is higher after the tax change, C firms tend to increase more at the beginning and take longer to reach the new steady state. Entry rates decrease by a very small amount compared to that of exit rates, which is consistent with the insignificant empirical results for entries. The exit rate of C firms decreases more than that of NC firms. Since this is a partial equilibrium transition path, the important price adjustment channel can not be illustrated in this set of figures.

Table 8: *Channel Decomposition, Steady State: Changes*

	Direct tax effect (τ_1, p_0, M_0)		Entry mass adjustment (τ_1, p_0, M_1)		GE price adjustment (τ_1, p_1, M_1)		Total effects (τ_1, p_1, M_1)	
	C	NC	C	NC	C	NC	C	NC
GDP (% of baseline)	-0.898	-0.158	-4.923	-1.933	+4.818	+2.030	-1.003	-0.062
Employment (% of baseline)	-0.898	-0.158	-4.923	-1.933	+5.342	+2.224	-0.479	+0.133
Entry rate (pp)	+0.000	+0.003	-0.000	-0.000	-0.003	-0.038	-0.002	-0.035
Exit rate (pp)	+0.046	+0.013	-0.000	-0.000	-0.278	-0.176	-0.231	-0.163

To quantify the proposed two impact channels, which are direct tax impact and general equilibrium price adjustments, Table 8 displays the key dynamics in the economy by shutting off one channel at a time. Column 1 displays the direct tax effect, which compares the results with the new tax rate (τ_1) with the baseline while holding the price and entrant mass at the

Figure 6: Transition path (partial equilibrium)



baseline level. With this direct tax impact, both GDP and employment decrease, and the exit rates increase. The increase in exit rates is about 0.046 percentage points for C firms, while that is only 0.013 percentage points for NC firms. This can recover the observed large increase in exit rate in the short term, as in the empirical motivation. Column 2 shows the impact of entry mass adjustment. Compared with results in column 1, this allows the entrant mass to adjust. It is shown that, with a stationary distribution of firms, entry and exit rates do not change with this scaling impact, and the aggregate output and employment decrease as total mass drops. Column 3 shows the impact of general equilibrium price adjustment, where the prices are updated to the new steady state levels. With price increases, exit decreases as it is more profitable for production. Quantitatively, this general equilibrium effect outweighs the direct tax impact; thus, there is a decrease in exit rates in the long-term in the total effects.

5.4 Counterfactual: lower business entry barrier

This model can also provide evaluations of the effect of different entry barriers on business dynamism. This subsection provides an example of examining the impact of lower business barriers on firms' entry and exit decisions with a fixed tax regime.

There are ongoing debates about the impact of rapidly growing artificial intelligence (AI) on labor market and firm dynamics. As generative AI technologies have become more accessible and less costly, recent evidence suggests a notable increase in AI adoption among small businesses.

⁸ Lower cost of generative AI may reduce entry barriers by enabling entrepreneurs to form

⁸JPMorgan Chase uses transaction-level data of small businesses to study the use of AI among small businesses. It is shown that the share of businesses that adopt AI since the first month of starting operation jumped from 1.2% in 2019 to about 6.5%

one-person companies. For instance, generative AI tools could be very helpful in tasks such as market research, regulatory compliance, translation, and market strategy development.

However, it is hard to quantify the extent to which AI helps lower the business entry barriers. One possible proxy for the reduction in entry costs can be defined as the median AI spending. Based on the transaction-level data for small businesses, the initial AI adopter's median AI spending decreases by about 60% from about \$50 in 2019 to about \$20 in 2024, as reported by JPMorgan Chase. Combined with the previous observation that an average of 3.85% of small businesses adopt AI in the first month of entering the market, it is reasonable to assume that the development of generative AI could have a 2.3% decrease in entry cost. While this 2.3% decrease in AI adoption cost does not fully capture the impact of AI on entry cost, it could be a reasonable starting point. So, in this exercise, I model a 2.3% reduction in entry costs for both firm types and evaluate the impact on firm dynamics.

Table 9: Entry and Exit (Policy 2 vs. Baseline)

	C Sector	NC Sector	Total
<i>Panel A: Masses (% change)</i>			
Total firms	+2.491	+1.228	+1.710
Continuing firms	+2.134	+1.018	+1.458
Entrants	+6.374	+2.573	+3.629
Exiters (all)	+6.374	+2.573	+3.629
<i>Panel B: Rates (pp change)</i>			
Entry rate	+0.003	+0.038	+0.017
Exit rate	+0.320	+0.180	+0.219
Entrant exit rate	+0.317	+0.141	+0.202

Table 9 shows that the total number of firms increases for both firm types. This is because more potential entrants find it profitable to enter the market as the entry cost decreases. The increase in the number of C firms is about twice as large as that for NC firms, as they are more bounded by the non-negative profits assumption with the larger up-front investment requirement. Similar to the case of tax changes, the entry rates barely change compared to the changes in exit rates. In general, lower entry cost boosts business dynamics in the economy.

Table 10: Equilibrium Outcomes (Policy 2 vs. Baseline, % change)

	C Sector	NC Sector	Total
Relative price (p_C/p_{NC})	-0.233		
Entrant mass	+6.374	+2.573	+3.629
Output	+0.871	+0.146	
Nominal GDP	+0.636	+0.146	+0.400
Average firm size	-1.759	-1.157	-1.374
Avg productivity, all operating firms	+0.015	+0.123	+0.065
Avg productivity, incumbents (operating)	+0.012	+0.114	+0.058
Employment	+0.337	-0.152	+0.064
Consumption	+0.757	+0.079	
Aggregate consumption (CES)	+0.413		
Wage	+0.298		
Real wage	+0.413		

With the general equilibrium effect shown in Table 10, an increase in firm counts creates higher labor demand that pushes up wages and lowers the relative price of C goods. On the in 2025. See <https://www.jpmorganchase.com/institute/all-topics/business-growth-and-entrepreneurship/understanding-ai-use-by-small-businessesfootnote-14>.

aggregate level, Table 10 shows that both nominal and real output increase. Average productivity increases, especially for NC firms. So, this could provide illustrative evidence that if AI could substantially lower the entry cost for firms, aggregate productivity could increase as business dynamism increases. The increase in average productivity is larger for NC firms as they are relatively smaller in scale. Total employment increases mildly with an increase in the C sector but decreases in the NC sector. Total consumption of the two goods increases.

6 Conclusion

The paper provides new evidence on how corporate taxation shapes labor market outcomes and the aggregate economy through firm dynamics. Because entry and exit are closely tied to job creation and destruction, understanding their responsiveness to tax policy is crucial for designing efficient and equitable fiscal systems. The study also provides additional evidence that business dynamism is closely tied to aggregate output and productivity. The analyses clarify that capital-intensive sectors, such as manufacturing and infrastructure, face disproportionate tax burdens that may unintentionally hinder employment in capital-intensive industries. Empirically, the paper finds that a one-percentage-point increase in the local trade tax rate is associated with a 0.8-percentage-point increase in exit probability among capital-intensive establishments, compared with 0.4 percentage point for establishments overall. In the calibrated model, a one-percentage-point tax increase reduces the total mass of firms, with a substantially larger decline among capital-intensive firms. The tax increase also reallocates employment across sectors and lowers productivity with a less active market. Together, these findings show that capital intensity strongly shapes how tax and entry-cost policies affect business formation, sectoral employment, and aggregate economic outcomes. This project offers a comprehensive framework for understanding how variations in corporate taxation translate into sectoral employment patterns and long-run productivity.

References

- B. Y. Aw, S. Chung, and M. J. Roberts. Productivity, output, and failure: a comparison of taiwanese and korean manufacturers. *The Economic Journal*, 113(491):F485–F510, 2003.
- R. J. Barro and J. Furman. Macroeconomic effects of the 2017 tax reform. *Brookings papers on economic activity*, 2018(1):257–345, 2018.
- E. Bartelsman, J. Haltiwanger, and S. Scarpetta. Cross-country differences in productivity: The role of allocation and selection. *American economic review*, 103(1):305–334, 2013.
- S. Basu and J. G. Fernald. Returns to scale in us production: Estimates and implications. *Journal of political economy*, 105(2):249–283, 1997.
- L. Bellmann, M. Gensicke, K. Grau, S. Kohaut, I. Möller, M. Oberfichtner, B. Schwengler, N. Tschersich, and M. Umkehrer. Iab establishment panel (iab-bp) 1993-2024. Technical report, Institut für Arbeitsmarkt-und Berufsforschung (IAB), Nürnberg [Institute for ... , 2026.
- D. Berger, K. Herkenhoff, and S. Mongey. Labor market power. *American Economic Review*, 112(4):1147–1193, 2022.
- C. Broda and D. E. Weinstein. Globalization and the gains from variety. *The Quarterly journal of economics*, 121(2):541–585, 2006.
- G. L. Clementi and B. Palazzo. Entry, exit, firm dynamics, and aggregate fluctuations. *American Economic Journal: Macroeconomics*, 8(3):1–41, 2016.
- S. Crawford and G. Markarian. The effect of the tax cuts and jobs act of 2017 on corporate investment. *Journal of Corporate Finance*, 87:102619, 2024.
- E. M. Curtis, D. G. Garrett, E. C. Ohn, K. A. Roberts, and J. C. S. Serrato. Capital investment and labor demand. Technical report, National Bureau of Economic Research, 2021.
- A. Ganzer, A. Schmucker, and S. Wolter. Establishment history panel 1975-2023. Technical report, Institut für Arbeitsmarkt-und Berufsforschung (IAB), Nürnberg [Institute for ... , 2024.
- D. G. Garrett, E. Ohn, and J. C. Suárez Serrato. Tax policy and local labor market behavior. *American Economic Review: Insights*, 2(1):83–100, 2020.
- X. Giroud and J. Rauh. State taxation and the reallocation of business activity: Evidence from establishment-level data. *Journal of Political Economy*, 127(3):1262–1316, 2019.
- T. Hethey-Maier and J. F. Schmieder. Does the use of worker flows improve the analysis of establishment turnover? evidence from german administrative data. *Journal of Contextual Economics–Schmollers Jahrbuch*, 133(4):477–510, 2013.
- H. A. Hopenhayn. Entry, exit, and firm dynamics in long run equilibrium. *Econometrica: Journal of the Econometric Society*, pages 1127–1150, 1992.
- A. Ljungqvist and M. Smolyansky. To cut or not to cut? on the impact of corporate taxes on employment and income. Technical report, National Bureau of Economic Research, 2014.
- P. McAdam, P. Meinen, C. Papageorgiou, and P. Schulte. *Returns to scale: New evidence from administrative firm-level data*. Number 24/2024. Deutsche Bundesbank Discussion Paper, 2024.

- P. Sedláček and V. Sterk. Reviving american entrepreneurship? tax reform and business dynamism. *Journal of Monetary Economics*, 105:94–108, 2019.
- E. Zwick and J. Mahon. Tax policy and heterogeneous investment behavior. *American Economic Review*, 107(1):217–248, 2017.

A Detailed Industry List

Table A.1: Industry Classification by Capital Intensity

NACE Rev. 2	Industry
<i>Panel A: Non-Capital-Intensive (NC) Industries</i>	
I	Accommodation and food service activities
S	Other service activities
F	Construction
G	Wholesale and retail trade; repair of motor vehicles and motorcycles
Q	Human health and social work activities
K	Financial and insurance activities
M	Professional, scientific, and technical activities
P	Education
O	Public administration and defence; compulsory social security
<i>Panel B: Capital-Intensive (C) Industries</i>	
A	Agriculture, forestry, and fishing
R	Arts, entertainment, and recreation
C	Manufacturing
H	Transportation and storage
N	Administrative and support service activities
B	Mining and quarrying
J	Information and communication
E	Water supply; sewerage, waste management, and remediation activities
D	Electricity, gas, steam, and air conditioning supply
L	Real estate activities

B Robustness checks for empirical motivations

In 2008, Germany's Business Tax Reform Act of 2008 lowered the business tax rate at the federal level and lowered the tax base for the local trade tax rate. The results in the main specification control for the impact of the 2008 reform with year fixed effects. Table B.1 provides a robustness check of the results by running the same specifications as in Table 1 but excluding the sample in 2008. The results that there are larger point estimates for capital-intensive establishments than those for all samples still hold.

Table B.1: Impact of trade tax change on exits excluding year 2008

	(1) All	(2) Capital intensive	(3) Small	(4) C and Small	(5) All	(6) Capital intensive	(7) Small	(8) C and Small
Tax changes	0.002** (0.001)	0.006** (0.003)	0.003** (0.001)	0.007** (0.003)	0.003** (0.001)	0.007** (0.003)	0.003** (0.001)	0.007** (0.003)
Tax changes, lag 1	0.001 (0.001)	0.001 (0.002)	0.001 (0.001)	0.002 (0.003)	0.001 (0.001)	0.001 (0.002)	0.001 (0.001)	0.002 (0.003)
Tax changes, lag 2	0.004** (0.002)	0.005** (0.002)	0.004** (0.002)	0.005** (0.003)	0.004** (0.002)	0.005** (0.002)	0.004** (0.002)	0.005** (0.003)
1 to 19	0.017*** (0.003)	0.022*** (0.005)			0.017*** (0.003)	0.022*** (0.005)		
20 to 199	0.003 (0.002)	0.005 (0.004)			0.003 (0.002)	0.005 (0.004)		
Age	-0.003*** (0.000)	-0.003*** (0.001)	-0.003*** (0.000)	-0.003*** (0.001)	-0.003*** (0.000)	-0.003*** (0.001)	-0.003*** (0.000)	-0.003*** (0.001)
Age ²	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
Log of Population	-0.058 (0.074)	-0.157 (0.118)	-0.066 (0.080)	-0.164 (0.135)	-0.056 (0.074)	-0.162 (0.117)		
Individual company	0.013*** (0.004)	0.021** (0.010)	0.013*** (0.004)	0.019 (0.013)	0.012*** (0.004)	0.021** (0.010)	0.012*** (0.004)	0.019 (0.013)
Partnership	0.002 (0.005)	0.009 (0.012)	0.003 (0.006)	0.007 (0.015)	0.002 (0.005)	0.009 (0.012)	0.002 (0.006)	0.007 (0.015)
Corporation	-0.003 (0.003)	0.005 (0.009)	-0.004 (0.004)	0.001 (0.013)	-0.003 (0.003)	0.005 (0.009)	-0.004 (0.004)	0.001 (0.013)
Mean exit rate	0.027	0.025	0.037	0.036	0.027	0.025	0.037	0.036
Short controls	Yes	Yes	Yes	Yes	No	No	No	No
Long controls	No	No	No	No	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	No	Yes	No	Yes	No	Yes	No
Year x Sector FE	Yes	No	Yes	No	Yes	No	Yes	No
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of Obs.	70,435	29,191	47,682	18,069	69,010	28,610	46,804	17,727
R-Squared	0.024	0.049	0.024	0.053	0.024	0.049	0.025	0.053

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors are clustered at the district level. The sample period is from 2008 to 2017. Regressions are adjusted using panel weights.

C Tax sensitivity analysis

These figures display changes in relative prices of two goods and the endogenously determined entrant mass changes over a span of tax rates. The dotted gray line is the baseline tax rate level. The sensitivity analysis shows that the impact of tax rate changes on relative prices is not linear. As the tax rate increases, the gaps between the two types of firms become larger.

Figure C.1: Tax sensitivity analysis

